SUPPLY CHAIN ANALYTICS REPORT

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1. EXECUTIVE SUMMARY

This report presents a comprehensive analysis of supply chain operations based on transactional data consisting of 12,000 orders. The objective of this study is to evaluate operational efficiency, delivery performance, product contribution, supplier effectiveness, and customer satisfaction.

The analysis identifies key inefficiencies in delivery operations, highlights revenue concentration across products, and establishes a clear relationship between operational performance and customer outcomes. Advanced KPIs such as OTIF (On-Time In-Full), return rate, and delivery efficiency metrics are used to assess system performance.

The findings indicate that supply chain inefficiencies are multi-dimensional, involving delivery delays, supplier variability, warehouse inconsistencies, and moderate customer satisfaction levels.

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2. OBJECTIVES OF THE ANALYSIS

The primary objectives of this analysis are:

- To evaluate overall supply chain performance and delivery efficiency
- To identify key drivers of cancellations, delays, and returns
- To analyze product-level contribution to revenue and demand
- To assess supplier and warehouse operational performance
- To understand customer satisfaction and return behaviour
- To design actionable KPIs for performance monitoring

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3. DATASET OVERVIEW

The dataset contains 12,000 records with 16 attributes representing supply chain transactions.

Key Variables:

- Order Information: order_id, product, quantity
- Location Data: city, warehouse, supplier
- Financial Metrics: price, discount, transport_cost
- Time Variables: order_date, delivery_date
- Performance Indicators: delivery_status, returned, customer_rating
- Additional Fields: payment_method, notes

Data Characteristics:

- Mixed data types (categorical and numerical)

- Missing values in customer_rating (~17%)
- Presence of inconsistent entries such as “Unknown”
- Unstructured textual data in notes field

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4. DATA PREPROCESSING AND CLEANING

The dataset underwent multiple preprocessing steps to ensure analytical accuracy:

- Conversion of order_date and delivery_date into datetime format
- Identification and handling of missing values in customer_rating
- Standardization of categorical variables such as supplier and warehouse
- Treatment of inconsistent values (e.g., “Unknown” entries)
- Validation of logical relationships between order and delivery dates

These steps ensured the dataset was reliable for downstream analysis and KPI computation.

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5. KEY PERFORMANCE INDICATORS (KPIs)

To evaluate supply chain efficiency, the following KPIs were defined:

5.1 OTIF (On-Time In-Full)

Measures the percentage of orders delivered successfully within the defined time threshold (7 days).

OTIF % = (Orders Delivered On Time / Total Orders) × 100

5.2 Delivery Success Rate

Measures the proportion of successfully delivered orders.

Delivery Success Rate = Delivered Orders / Total Orders

5.3 Return Rate

Indicates the percentage of orders returned by customers.

Return Rate = Returned Orders / Total Orders

5.4 Average Delivery Time

Measures average number of days taken to deliver orders.

5.5 Revenue

Revenue is calculated as:

Revenue = (Price × Quantity) – Discount

5.6 Delivered to Returned Ratio

Represents the number of successful deliveries for every returned order, providing a direct measure of fulfillment quality.

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6. EXPLORATORY DATA ANALYSIS AND INSIGHTS

6.1 Delivery Performance Analysis

Delivery performance shows significant variability across the dataset. Orders are categorized into Delivered, Cancelled, and Pending, indicating that fulfilment is not consistently successful.

Analysis reveals that:

- A portion of orders fail to reach completion
- Delivery delays contribute to inefficiencies
- Operational issues are concentrated in specific segments

This highlights the need for improved logistics coordination.

6.2 Product and Revenue Analysis

Product-level analysis indicates:

- High variability in product pricing (₹1000 to ₹50,000)
- Uneven distribution of order quantities
- Revenue concentration among a limited set of products

This suggests a Pareto-like pattern where a small subset of products contributes disproportionately to total revenue.

6.3 Supplier and Warehouse Performance

Supplier analysis reveals inconsistencies in delivery performance:

- Certain suppliers exhibit higher cancellation and return rates
- Supplier performance is not uniform across the dataset

Warehouse analysis indicates:

- Uneven load distribution

- Presence of “Unknown” warehouse entries

These findings point to gaps in operational tracking and resource allocation.

6.4 Customer Experience Analysis

Customer satisfaction is moderate, with an average rating of approximately 3 out of 5.

Key observations:

- Missing rating values (~17%) indicate incomplete feedback capture
- Lower ratings are associated with higher return rates
- Customer experience is directly impacted by delivery performance

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6.5 Return Behaviour Analysis

Returns serve as a critical indicator of supply chain inefficiency.

Findings show:

- Returns are linked to both product issues and delivery failures
- Certain products exhibit higher return frequencies
- Operational inconsistencies contribute to increased return rates

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7. BUSINESS INSIGHTS

- Supply chain inefficiency is multi-factor driven, involving delivery delays, supplier variability, and warehouse inconsistencies
- Revenue is highly concentrated among a limited number of products
- Customer satisfaction remains moderate and is impacted by operational performance
- Returns are a strong indicator of service and product quality issues
- Data quality issues (missing ratings, unknown warehouse entries) affect analytical depth

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8. BUSINESS RECOMMENDATIONS

- Improve delivery tracking and enforce strict SLA adherence
- Optimize supplier performance through monitoring and benchmarking
- Eliminate “Unknown” warehouse entries to improve operational visibility
- Strengthen customer feedback mechanisms to improve data completeness
- Focus on high-performing products while optimizing underperforming ones

- Implement continuous KPI monitoring using dashboards

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9. DASHBOARD IMPLEMENTATION

A Power BI dashboard was developed to visualize insights and monitor KPIs across five key areas:

1. Executive Overview
2. Delivery Performance
3. Product & Revenue Analysis
4. Supplier & Warehouse Performance
5. Customer Experience

The dashboard enables real-time tracking and supports data-driven decision-making.

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10. CONCLUSION

This analysis demonstrates how supply chain data can be transformed into actionable insights through structured data cleaning, exploratory analysis, and KPI development.

The study identifies key inefficiencies in delivery operations, highlights revenue concentration risks, and establishes a clear relationship between operational performance and customer satisfaction.

By implementing the recommended improvements and leveraging KPI-based monitoring, organizations can enhance efficiency, reduce operational risks, and improve overall customer experience.

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11. FUTURE SCOPE

- Implementation of predictive models for delivery delays
- Supplier performance scoring systems
- Real-time KPI dashboards
- Integration with inventory and demand forecasting systems